ETSN11, Final exam 2023-10-25, Solutions

This document includes our proposed solutions. The solutions are quite short and may not have given full points on the exam. Other solutions, which we have not thought about, can also be correct. **General advice**: In **Part A,** you need to correctly describe the specific terms and concepts to get points. In **Part B**, you need to motivate your chosen formulas and calculations to get full points. In **Part C**, you need to give a suitable solution to the specific scenario. This solution must address the requirements and challenges in the scenario, and include specific protocols and technologies in order to get full points.

Part A: Technologies and architectures (30 points)

In the multiple-choice questions, you get points for each correct answer, and point reductions for each incorrect answer. You cannot get a lower score than 0 in the multiple choice questions.

1. Which of the following statements are true for a **Decentralized network architecture**? **(3p)**

a. This architecture can be used for time-critical applications.
b. This architecture can be used for non-critical applications.
c. "Decentralized" means that there is no gateway in the network.
d. "Decentralized" means that the nodes organizes themselves.
e. "Decentralized" means that the nodes are distributed over a larger area.
f. LoRa uses a decentralized network architecture.
g. ZigBee uses a decentralized network architecture.

2. Which of the following statements are true for WiFi 6 mesh? **(3p)**

a. It is a network technology aimed at time-critical applications.
b. It is a network technology aimed at non-critical applications.
c. It uses a centralized network architecture.
d. It uses a decentralized network architecture.
e. It is good for resource-constrained devices.
f. It uses delay-tolerant networking.
g. Animal and environmental monitoring are typical applications for this technology.
h. Industrial automation and robotics are typical applications for this technology.

3. Which of the following statements are true for **NB-IoT**?? **(3p)**

a. It is a good technology for time-critical applications.
b. It is a good technology for resource-constrained networking.
c. It is a good technology for underwater networks.
d. A device must connect with a base-station before it can send data.
e. A device can send data at any time (i.e ad-hoc).
f. It is a single-hop network technology.
g. It is a multi-hop network technology.

4. Which of the following statements are true? **(3p)**

a. Underwater communications can use radio, optical and acoustic signals.
b. Interplanetary communication can use radio, optical, and acoustic signals.
c. The Licklider transmission protocol does not use ACKs.
d. The Licklider transmission protocol uses ACKs for some of the data.
e. The Licklider transmission protocol uses ACKs for all data.
f. A DTN is based on specific network protocols that handle store-and-forward data dissemination.
g. A DTN is based on specific link protocols that handle store-and-forward data dissemination.
h. A DTN is based on specific application protocols that handle store-and-forward data dissemination.

5. Which of the following statements are true?? **(3p)**

a. Starlink is a good technology for time-critical applications.
b. Argos is a good technology for time-critical applications.
c. Starlink is a good technology for non-critical applications.
d. Argos is a good technology for non-critical applications.
e. Argos provides full duplex communication.
f. Starlink provides full duplex communication.
g. Argos provides one-way communication.
h. Starlink provides one-way communication.
i. Argos provides half-duplex communication.
j. Starlink provides half-duplex communication.

6. Describe the general architecture of a **sensor network** and give an example of an application that uses a sensor network? **(3p)**

Solution: A sensor network consists of sensor nodes and one or more sink/gateway nodes which connects to the back-end via the Internet. The sensor nodes collect data and transmit this data to a sink, which then forwards the data to the back-end. The communication can be single-hop or multi-hop and the architecture can be centralized or decentralized. One typical application is animal monitoring, where sensors are placed on animals. Other applications can be environmental monitoring and industrial systems.

7. What is **fog computing**? **(2p)**

Solution: In fog computing, cloud computing resources are distributed and placed closer to the users. This reduces computing latencies. **Note**: CDN is not fog computing.

8. Describe the purpose and general architecture of a **Private 5G Network**. **(3p)**

Solution: In a Private 5G network, an organization can get reserved 5G capacity. A Private 5G network can be deployed in two ways: (1) By using Network slicing, an operator can reserve capacity in a public 5G network for the organization; (2) The organization can build its own 5G network. This requires permission to use a licensed frequency band. **Note**: Private 5G network is a specific concept, as described above. If your answer does not show that you understand what a Private 5G network is, you will not be given full points.

9. Describe a **routing algorithm** that can be suitable for a **vehicular ad-hoc network** that should warn drivers about hazardous roads ahead. **(3p)**

Solution: There are several routing protocols that you can describe here (choose one and **describe** it). **Geocast** is a routing protocol that broadcasts messages in a zone-of-relevance, so vehicles that enter the hazardous area (for example after an accident), can forward messages to all vehicles behind them in a certain area. **GPSR** is a position-based routing protocol that sends messages in unicast to a user at a specific position. This requires that all vehicles know about other vehicles' positions. **Broadcast** can of course be used, where a warning message is sent to ALL other vehicles. **AODV** can be used, but is probably not that suitable since it does not target positions, only destination addresses. **Note**: **Gossiping** does not send messages to ALL nodes, because then it is broadcast.

10. Explain what the procedure **frequency hopping** is, and explain why it is used in communication systems. **(4p)**

Solution: Frequency hopping is a communication procedure where two nodes change carrier frequency several times during the transmission of one single message. The frequency changes are performed using a specific pattern that only sender and receiver knows about. This can reduce interference for license-free bands (used in Bluetooth), and it can also make it more difficult for another node to listen to the communication (therefore used in military systems). **Note**: Frequency hopping is a specific concept, which means that if you describe something else, this is not a correct answer.

Part B: Performance analysis problems (30 points)

11. You are doing performance modeling of a web server system. You measure the arrival rate of new requests and find that the arrivals can be modelled as a Poisson process with an average rate of 10 requests per second. Also, you measure the service times of request, and find that the service times are independent and that they can be modelled with an exponential distribution with an average of 0.06 seconds. Also, you see that the web server system can be modelled as a conservative single server system with unlimited queue. **(10p)**

a. What is the **average utilization** of the server? (2p)

Solution: The system as described above can be modelled as an M/M/1 system with $\lambda = 10$ requests per second and $\mu = 1/0.06 = 16.67$ requests per second.

The utilization = $\rho = \lambda/\mu = 0.6$.

b. What is the **average throughput** of the server system? (2p)

Solution: The throughput = $\lambda_{eff} = \lambda = 10$ requests per second (since there is no blocking in the system).

c. What is the **average total number of requests** in the server system? (2p)

Solution: For M/M/1 systems: $\overline{N} = \frac{\rho}{1-\rho} = \frac{0.6}{1-0.6} = 1.5$

d. Assume instead that the service times can be modelled as a **deterministic distribution**. What is the **average total number of requests** in the system in this case? (4p)

Solution: In this case, we can model the system as an M/D/1 system. For this system, the Pollaczhek-Kintchine formula is valid. For deterministic service times, $V[X] = E[X^2]-E[X]^2 = 0$, which means that $E[X^2]=E[X]^2 = 0.06^2 = 0.0036$.

This means that $\overline{N} = \rho + \frac{\lambda^2 \cdot E[X^2]}{2(1-\rho)} = 1.05$

12. You are modelling a very simple MAC protocol. A device can send a data packet at any time. If a request is received correctly by the receiver, the receiver sends an ACK packet after exactly 1 second. If the device does not receive the ACK packet correctly, it retransmits the data packet after another 1 second. The cycle repeats until the device has received an ACK. The probability that any data or ACK packet are received correctly is 0.8. **(8p)**

What is the average time from when the device starts to send its data packet until it has received an ACK?

In your calculations, you can ignore all processing, transmission, and propagation delays. Also, the devices are perfectly synchronized.

Solution:

P(Packet received correctly (Data and ACK)) = 0.8

A successful transmission = Data received correctly **AND** ACK received correctly.

P(a successful transmission) = P_S = P(Data received correctly) * P(ACK received correctly) = $0.8^2 = 0.64$.

P(a failed transmission) = $1 - P_S$.

P(*k* transmission cycles for a successful transmission) = $= P_S \cdot (1 - P_S)^{k-1}$ (geometric distribution)

E(Number of transmission cycles for a successful transmission) = $1/P_S = 1.56$

Length of a failed transmission cycle = 2 seconds

Length of a successful transmission cycle = 1 second.

E(Time for a successful transmission) = $2 \cdot 1.56 - 1 = 2.1$ seconds (reduce with 1 second for the successful cycle).

13. You are designing a communication link from the Earth to the Moon. The sender is one of several ground stations on the Earth, and the receiver is a ground station on the Moon. You can assume that there always is line-of-sight between the two ground stations (you use the optimal ground station on the Earth for your data transmission). You achieve an average transmission capacity of 1 Mbits/second on your communication link. What will the average total **one-way delay** be for a data message of **10 kbits**? **(4p)**

Solution:

Total delay = Transmission delay + Propagation delay

Transmission delay = (Number of bits) / (Transmission speed)

Transmission delay = $\frac{10 \cdot 10^3}{10^6}$ = 0.01 seconds

Propagation delay = (Propagation distance) / (Propagation speed)

Propagation distance = $384 \cdot 10^3$ km (distance to the Moon)

Propagation speed = $3 \cdot 10^8$ m/s (speed of light/radio)

Propagation delay = $\frac{384 \cdot 10^3 \cdot 10^3}{3 \cdot 10^8} = 1.28$ seconds

Total delay = 1.29 seconds

14. You are investigating the performance of the queueing system shown in the figure. New requests are arriving as a Poisson process with mean λ=6 requests per second. The service times are independent and exponentially distributed with mean service rate of $\mu = 5$ requests per second. The queue is limited with only 2 places. You perform measurements on the queuing system and find that at any random time instant, the probabilities that there are k requests in the system (k = 0 to 3) are: P(k=0) = 0.19; P(k=1) = 0.23; P(k=2) = 0.26; P(k=3) = 0.32. Also, a request is simply blocked if it arrives when the queue is full. The queueing policy is first-come-first-serve (i.e an ordinary queueing system) **(8p)**

a. What is the average number of requests in the system? (2p)

b. What is the average throughput of the system? (2p)

c. Assume that a request R arrives and takes the last place in the queue (i.e, there are 2 requests in the system when it arrives). What is the average total response time for request R? (4p)

Solution:

This is a single server queuing system with limited queue, so not one of the queuing models that we have seen in the course before. But, you have all information to solve all the questions with probability theory and statistics.

a. E[Number of jobs in the system] = $0 \cdot P(k=0) + 1 \cdot P(k=1) + 2 \cdot P(k=2) + 3 \cdot P(k=3) = 1.71$

b. The throughput is always the effective incoming arrival rate ("what goes in, comes out"), i.e λ_{eff}

In this case: $\lambda_{eff} = \lambda \cdot (1 - P(k=3)) =$ 4.08 requests/second

c. When the arriving request R takes the last place in the queue, there is one request (R_S) in the server, and one request in the queue (R_Q)).

The average response time for R = (The average remaining service time for R_S) + (The average service time for R_Q) + (The average service time for R)

The average service time for R = The average service time for R_Q = $1/\mu$ = 0.2 seconds.

The remaining service time for R_S is also called the residual service time for R_S, and it can cause a lot of mathematical problems in queuing theory. However, in this case, the service times are **exponentially distributed, i.e memoryless**, which means that the average residual time for R_S is the average service time = $1/\mu$.

Therefore, the average response time for R = $\frac{1}{\mu} + \frac{1}{\mu} + \frac{1}{\mu} = 0.6$ seconds.

If you have answered correctly, but without any motivations, you will not get any points on c, since the residual service time for the request in the server is important to take into account.

Part C: System design scenarios (30 points)

For each of the scenarios given below, there are some general recommendations for the system design in this scenario, and some solution examples, which were given full points on the exam (and some that are smart, but lack some details to get full points). The solutions have been only minor edited, fixing typos and grammatical errors.

15. The Skåne University Hospital (SUS) in Lund wants you to design a network system for them. The network system should track different types of mobile medical equipment, since medical equipment has a tendency to get lost in the large hospital. There are about 1000 medical equipment devices, placed on small carts with wheels, which should be tracked. You can focus on the 12 story central building, also called Blocket. SUS wants to monitor the location of the devices, and the most important requirement is to know on which story (floor) a specific device is placed. The image below shows a typical cart for medical equipment. **(10p)**

Clarification during the exam: Each medical device is placed on a cart, and will not change cart. So, it is enough to track carts.



The scenario: This is very much a real-life scenario. The hospital has large problems with lost medical equipment, and there have been projects with Lund University in order to solve this problem using different tracking solutions. In reality, a medical equipment does not have to be placed on a cart, and it is not enough to track the floor, so these two things are simplification of the reality to make the scenario easier to solve.

Recommendations: This is a non-critical monitoring application with battery-powered devices (resource-constrained networking), where you need to track the devices on several floors. From the course, you should know several suitable technologies, for example: LoRa, NB-IoT, and Zigbee. Use one of these technologies, and describe how you would deploy the network architecture in this specific scenario. **Note**: WiFi and 5G are not suitable technologies for resource-constrained networking. To suggest that the sensor devices recharge in power outlets is a rather major limitation of the scenario, since this solution requires that staff keep track of the carts and make sure that they re-charge. Further, GPS cannot be used here, both because it is indoor, and that there are multiple floors on top of each other (with the same GPS coordinates).

Solution example 1:

A solution that would work perfect for this system is 5G's mMMTC (massive machine type communication), which should handle large amounts of connected devices. The hospital could use network slicing of the 5G network as other services would not interfere. Since the hospital consists of 12 floors, one single radio station would probably not be enough. We can therefore use C-RAN (cloud radio access network) to have smaller nodes on each floor, where the main processing broadband unit could be placed on the main floor. The stations would be connected to the electricity grid.

Using NB-IoT we could let the carts have low cost sensors, that could transmit data on the deployed cellular network. It could transmit data when it has been moved, and some technology of knowing the story should be possible here. Since the sensor device must request to send data (using RACH - random access channel) it would certainly not be impossible to have the cellular node/station transmit information of the story it is installed on in this "handshake". The floor should be included in the data that is sent by the sensor. The data that is sent is extremely small since it only has to contain which floor, so it would not congest the network. The data could be sent to a server in the hospital, and the employees could have an application on their phone that could access this information.

The sensors are battery powered, and since they only transmit when they are moved this battery should be good for a very long time. A sensor could use some type of energy harvesting where movement derives electricity (like some clocks do) if the sensor battery is not considered enough.

Solution example 2:

This is a Non-critical application. The hospital only needs to know where the equipment is when it is lost, and they do not need to track its position in real-time. It is therefore okay with some latency. Also, for simple transmission of position-data, high data rates are not required. It is also preferable if this solution does not force the equipment carts to use high amounts of energy.

If I have understood the scenario correctly, it is the carts that need to be tracked and not each and every one of the 1000 medical equipment devices. So it is enough to know the position of the carts.

My solution is based on LP-PANs and involves setting up multiple ZigBee networks. On each floor there will be one ZigBee Coordinator. One of the characteristics of a Zigbee coordinator is that it contains "information" about the network. Among this information is a network ID and since the coordinators will not move once they are installed, this ID is easily mapped to a certain story in the building.

The medical equipment carts are equipped with ZigBee End devices. These occasionally send a message to the coordinator. In this way, a coordinator will know which devices

are in its network (on its floor) through the unique device ID. If we decide that it is sufficient to know on what floor a cart is, a very simple message (simply announcing its presence) will be enough. The range of communication between two Zigbee devices is approximately 100m. Since there could be times when a cart (end device) is further than 100 m away from the Zigbee coordinator, Zigbee routers will also be placed on each floor (probably one or two, depending on the surface area of each floor). The Zigbee routers serve as intermediaries, extending the range of the communication beyond just coordinator-to-end device.

The coordinators send information about which Zigbee end devices they have in their Zigbee network to a server and from this server, the information is made accessible to the hospital staff. The server can be placed in the hospital and since they most likely have a Wi-Fi network in place already, the information can actually be sent from the coordinators over Wi-Fi. I believe it is fair to assume that a Wi-Fi network (with access points placed such that the entire building is covered) is in place, since it is necessary for the staff and also a nice-to-have thing for the patients.

This solution assumes that it is sufficient to know which floor a cart is on. If more detailed positioning is necessary, this will need to be solved in some way. GPS is not accurate enough to distinguish between stories in a building. It is possible that we could use some sort of beacon-based positioning system.

The carts (end devices) will be battery powered. End devices do not use a lot of energy. The routers and coordinators are stationary and can be mains-powered.

Solution example 3:

This is a typical non-critical, non-time-sensitive application. There is additionally a large number of devices, in a constrained area, with an additional requirement for low power usage (since they will need to be battery-powered). There is additionally a requirement for the devices to be position tracked.

A reasonable choice for this application would be 5G mMTC, optionally using a private 5G network if there is not already good enough coverage of the hospital from the public network. However, I feel that this would become too expensive for the hospital in the long run.

Therefore, I suggest using a LoRa network, where each sensor is a node. There could be one or more gateways/base stations on each floor, depending on how large each floor is. The sensors periodically transmit messages to the base stations, and after receiving a confirmation message in one of the two slots can go back to sleep, prolonging their battery life. I suggest that they only have to transmit the messages once every half hour, which gives a good overview of where they are located without drawing too much power. The sensors themselves have no built in positioning capabilities, but by designing the network so that each cart transmits to the closest base station (decidable for example by signal strength), only base stations on the current floor will receive the

message. The base stations in turn know their position and add this information when relaying the messages. The sensors are battery powered.

I am proposing a mesh network using single-hop, in a tree topology. It is unnecessary for the sensors to be able to relay messages between each other, because it is a constrained area and they will probably be in base-station reach all the time anyways. Additionally, multi-hop relaying would decrease battery life.

The base stations are suitably connected to a base server somewhere in the building through a backhaul such as existing Ethernet. The base server is responsible for collecting messages from all the base stations and presenting the information to the end users.

Solution example 4 (This solution is very smart, but it is lacking some details, so it was nog given full points)

To me this seems like a suitable application for a WSN network, as the required data is extremely limited, it is a non-critical application and we are practically using it as a monitoring system - which is a very common use case for WSNs.

- There is extremely little data to be sent to/from each cart, meaning the end nodes can be very simple, cheap and energy efficient.
- Carts are easily accessible, thus the batteries will be easy to recharge/swap (even though this probably wont be needed too often due to the low energy use).
- Distances are relatively short, as it is within a building. However, there are walls and complex medical equipment in the way, so there will be some obstruction.
- There is likely already both cellular coverage and WiFi in place.
- Each cart needs to be uniquely identifiable

Medical carts are very unpractical to carry down the stairs, so I am making the assumption that they will all be traveling by elevator when changing floors. Therefore, I suggest placing central nodes near the elevator(s) on each floor, and placing an incredibly simple tag on each of the carts that can be detected by these nodes when they are nearby. As we are only interested in what floor the cart is on, I suggest an opportunistic routing scheme where the cart-tag conveys to any nearby elevator node “I am here". This node in turn forwards this information via the existing WiFi to a central database that stores the current/latest floor of each cart. By using this approach, we avoid the obstacle of obstruction from any medical devices/machines/thick walls etc, as we are only communicating when close by.

16. Scania is a large Swedish company and a world-leading truck manufacturer. **(10p)** Currently, Scania investigates solutions for so called vehicle platooning, that is, solutions where several self-driving trucks cooperate, thereby allowing them to drive with very small distance between each other, see an illustration in the image below. A vehicle platoon is always controlled by the leader vehicle, and other vehicles can "connect" behind the leader. Your task is to help Scania with a network system solution for their vehicle platoons. The company will then perform test drives in a secured test area outside of the city of Södertälje (south of Stockholm). The vehicle platoon should have one leader vehicle, and maximum four following vehicles.



The scenario: As you understand from the image, this is a real-life scenario. Scania and other truck manufacturer develop and test solutions for vehicle platooning, since platooning will have a major effect on the energy consumption for the following vehicles. Vehicle platooning can be deployed both with and without vehicle communication systems, however in this course, we only discuss networked solutions.

Recommendations: This is a typical time-critical application, where latency requirements are strict. There are specific communication technology solutions for vehicular systems with high mobility, where we have discussed IEEE 802.11p (WiFi), C-V2X (Cellular), IEEE 802.11ad (mmWave) in the course. All these technologies allow for ad-hoc connections without an access point. Preferably, you select one of these technologies and propose a specific solution for this scenario. Other technologies can be used, depending on what you propose. However, irrespective of technology, your solution must incorporate the chosen technology in a correct way. **Note**: Wifi-direct and WiFi 6 mesh are not really developed for outdoor scenarios where the access point moves with high mobility. Ordinary 5G networks do not provide ad-hoc connections, and you cannot set up your own 5G network without frequency spectrum permissions. Gossiping does not forward messages to all nodes, since then it is called broadcast or flooding. Geocast is problematic to use when the ZOR (zone-of-relevance) changes all the time. To propose a VANET without specifying the communication technology will give points reductions.

Solution example 1:

This is a time-critical application. Low latency is crucial (as it is a matter of road safety). High data rates are also required. Trucks have (essentially) unlimited power supply that can facilitate this as well as equipment for handling heavy computation related to this scenario.

The main idea here is a Vehicular ad-hoc Network (a VANET). There are two options to choose from when it comes to VANETs: IEEE 802.11p or Cellular - Vehicular-to-Everything (C-V2X).

For this scenario, I will use C-V2X with 5G NR. With 5G NR we can achieve very low latency and high reliability. The trucks will run an application that is able to dynamically determine and identify a leader truck. Once a truck has joined the platoon, all communication (such as instructions from the leader) will be done over 5G cellular network. GPS is used for rough positioning of the trucks and high precision positioning (regarding distances between trucks) can be accomplished using optics. The leader determines what distances trucks should keep to the truck in front and informs the other trucks of this.

This approach assumes good 5G cellular coverage which I believe there is in Södertälje (as it is an urban area, and close to Stockholm, which is a large city).

Solution example 2:

Requirements of the system:

- Self-driving trucks, in small groups, should be able to drive together in groups
- In every group, there should be one leader and at most four following vehicles

Challenges:

- Trucks are constantly moving
- Trucks usually drive over a long distance, which our solution needs to take into account
- Time-critical application

Quality of Service (QoS) requirements:

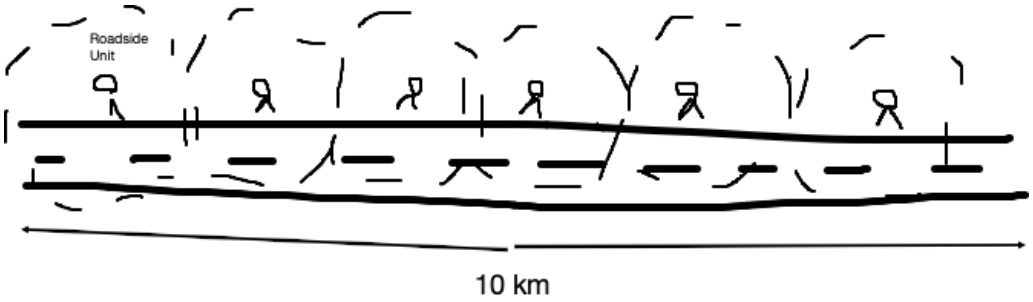
- Low latency
- High reliability

Quality of Experience (QoE) requirements:

- The truck platoon should remain intact and close to each other

As written in the requirements, our solution needs to take into account the long distances that trucks may drive. We want our solution to work beyond just the test area, so our solution cannot depend on location-specific technology. We could use Vehicular-to-Everything (V2X), which there is a standard for called IEEE 802.11p, for communication between the leader of the platoon and small roadside communication devices. The range of this protocol is around 1000 m, meaning that if we want to cover a

road of 10 km we would need to use maybe 6 or 7 devices, to have a margin of safety from their max range.



The picture above shows an example of how roadside units would be placed across a road. The routing protocol would be similar to cluster based routing, as the platoon leader would be like the cluster head node and would handle the communication with the roadside units and the following vehicles. Because we have the roadside units, we can put processing units in them to setup an edge computing system. The processing units could perform the calculations necessary for the platoon leader's self-driving, and then the leader can, using this information, send instructions to the following vehicles. This solution means that we would have very low latency. This, since the communication between the leader and the roadside unit happens over a short distance, and the data doesn't have to be first sent back to a central server and back for calculations, and the short communication distance also improves reliability of the communication, both of which are important QoS requirements. However, the roadside units still should send data back to a central server so that Scania can keep track of the location of a truck convoy.

The roadside units could be battery powered and have small solar panels to recharge their battery. Because they will be close to the road and roads usually are clear from trees and other obstructions for solar power, this will be able to keep them up and running. As a back-haul, they would use cellular to send data to the back-end. Roadside units that have a cellular connection will act as sinks and the roadside units that haven't any cellular connection would have a routing table of which roadside unit to forward to reach a sink. This can be hard-coded since these roadside units would be stationary.

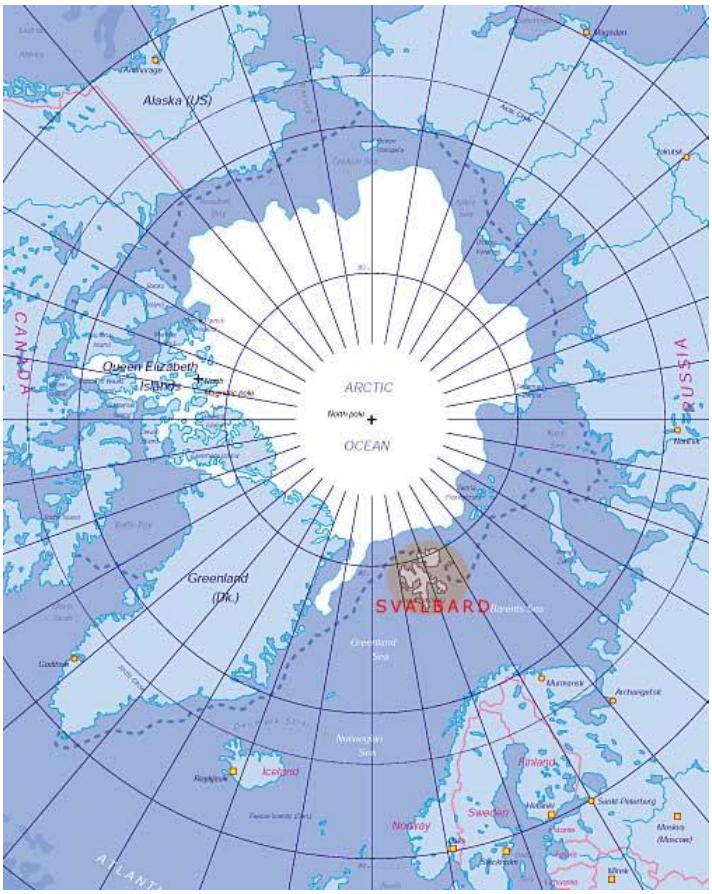
Solution example 3:

I assume that this solution will only be used in Sweden. Starlink has good coverage in whole Sweden, so we will have a satellite backhaul by using Starlink. The leader vehicle could have an Internet connection via a Starlink receiver/transmitter placed on the roof of the leader truck. There could be places with a lack of cellular coverage, which is the reason why Starlink is an appropriate approach here. There could be a hybrid solution where Cellular is used when available, and Starlink is used otherwise, in order for the trucks to have internet access. I think that an Internet connection is needed in order to monitor the trucks and send commands to the trucks regarding for example paths to take to the destination or when to emergency stop.

IEEE 802.11p could be used for communication among the trucks. Each truck has a networking device, which wirelessly connects the four trucks together. We have an ad hoc network where each truck is a node, which can route data to the correct truck/destination. If a message from the leader truck needs to reach the last truck in the platoon, the intermediate trucks route this data via an ad-hoc connections using RF. Since the distance between trucks may vary, it is required to have an ad-hoc solution in which all trucks are connected. If the last three truck were to be directly connected to the leader truck only, connection may fail due to varying distances during the trip and due to the intermediate trucks being obstacles for a solid RF communication.

The networking devices can be powered by the trucks battery power, which is continuously being replenished during the trip.

17. Your task is to help the polar research station NIPR, placed in the small town of Ny-Ålesund in Svalbard. NIPR wishes to monitor the waters around North Pole, and in particular, monitor the North Pole ice cap. Since the North Pole covers a very large area, NIPR wishes to use the waters just north of Svalbard as a test site. Your task is to propose a suitable network architecture for this monitoring system. The system should continuously monitor the environment, and collect data of the ice cap, water temperatures and salinity. NIPR has access to a cloud server, where all data should be stored. Below is a map showing the location of Svalbard.



The scenario: Several international research organizations monitor the environment in the Arctic region, and several of these organizations work from Svalbard. NIPR exists and performs research in the polar region with activities at Svalbard, but this specific scenario project is fiction.

Recommendations: This is a complex non-critical IoT scenario, with underwater environmental monitoring in a remote region. NIPR has not specified what type of equipment you should use, so you are quite flexible in terms of potential solutions. Svalbard has both electricity and Internet access (people are living there and there is an airport). In order to get full points, you should give a suitable complete and coherent system architecture that captures the important things of the scenario. In your solution, you need sensors and gateways, and a data dissemination strategy. The sensors and gateways must have a power supply, and the gateways need a backhaul. The data dissemination strategy can use DTN, and in this case, you need to include mobile DTN nodes (with power supply). Also, you need to explain specifically how you will monitor the ice cap. If you use Argos as backhaul in gateways, you need to include arguments about the limited data rate in order to get full points, since Argos allows for a very limited data rate (it is developed for sensors, not gateways). There are seals in the Arctic region (http://icestories.exploratorium.edu/dispatches/big-ideas/arctic-seals/), but seals are mammals and do not live under the ice. Starlink is actually available at the North Pole (https://www.highnorthnews.com/en/elon-musks-starlink-expands-provide-coverage-much-arctic). You are not required to know this before the exam, so any arguments around the availability of Starlink at the North Pole will not give you any point reductions. Also, to have diesel generators on the North Pole may not be a recommended strategy, both for environmental issues, and for the cost of refueling. However, this depends on how you will use diesel generators, and your arguments for them.

Solution example 1:

This is an example of a case where data will need to be collected from very remote locations. The data that needs to be collected is assumed to be of low bit-rate and non-critical.

The hardware platform that I would propose for this system is a buoy with the needed equipment and sensors. Multiple buoys will be placed over the relevant area and they will be equipped with sensors able to monitor the water temperature and salinity. For collecting data on the ice caps, I propose that the buoys have a radar, which will be able to detect where the ice cap is at the moment and if that is changing. This radar data should be possible to distillate down to only the most necessary parts and thus being small enough in size for the proposed communication solution Argos.

To supply the buoys with electricity either solar or wave power could be harnessed from the surroundings. My recommendation is to put solar panels on the buoy combined with some equipment for harnessing wave energy in addition. This will provide extra energy

and keep the buoy functional even during the times of the year when the sun don't shine. Furthermore, the buoy should have some type of battery to store generated excess power for later use when needed.

Since the information gathered from the buoy is of low bit-rate and can be seen as non-critical, the Argos satellite system provides a great solution for collecting this data. This also because there is only need for one-way communication from the sensors to the back-end which is what Argos offers. Furthermore, with the data being collected from a remote location that can be assumed to have no existing infrastructure to rely on, satellite communication is the best option.

After the Argos satellites have collected the data from the sensors, they will send this data back to Earth to one of the stations able to receive data from the satellites. From there, the data can be transported via the ordinary commercial Internet access to the cloud server where the data is to be stored.

Solution example 2:

Requirements of the system:

- Monitor the environment, and collect data of the ice cap, water temperatures and salinity
- Get that data to a cloud server for storage

Challenges:

- Svalbard is a remote location with harsh weather conditions and unknown internet infrastructure
- Getting the data from the sensors to the cloud server
- Power management of sensors and other equipment

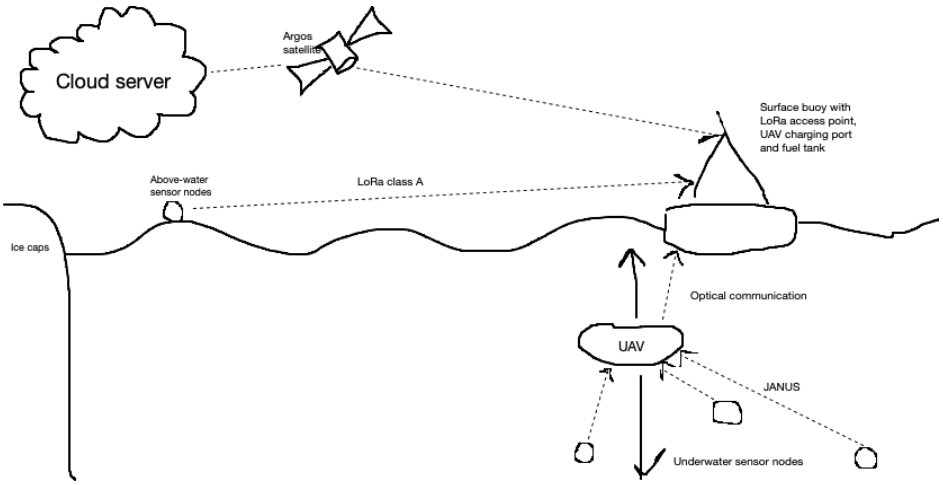
To collect data on the environment, ice caps and water temperatures and salinity, we are going to need a variety of sensors deployed in different locations to be able to gather all this data. To measure water temperature and salinity, we would need to spread out sensors across the area we want to monitor and keep them submerged to get our data. Then we would also need sensors near the ice caps to be able to collect that data.

To get the data to the cloud server, we could deploy surface buoys in the waters between Svalbard and the North Pole. These surface buoys could have Argos transmitters, which would transmit data to the Argos satellites, which then will send the data to our cloud server for storage. Argos is designed for environmental- and animal monitoring with resource-constrained devices, therefore the energy efficiency should be good for the surface buoys. The data they will transmit is the data that we get from the sensors. Because Argos has a data rate limit, we would need enough surface buoys spread across the area so that a single buoy doesn't have to upload more data than is allowed because many sensors send their data to it, and also so that the distance to the sensors won't be too large.

Assuming the sensors collecting data from the ice caps are above water, we could use LoRa Class A devices, and have a LoRa access point on the surface buoys. LoRa is designed for long range and low battery use, which means that we can monitor ice caps even if they are a distance away from the buoys, and we won't have to replace the batteries often on the sensors. Also, the ice cap sensors won't have high bit rate transmissions, and their data doesn't require guarantees of low latencies, so using LoRa which doesn't work with high bit rates, and with a contention-based MAC which class A devices use is appropriate. If we have other devices above water aside from the ones monitoring the ice caps, they will also be LoRa class A devices.

For the underwater sensors that measure the water temperature and salinity, we cannot use LoRa because of the attenuation of the signal. Instead, for these sensors we would transmit the data using a delay-and disruption tolerant network (DTN) protocol using underwater autonomous vehicles (UAVs). The UAVs would move slowly between the surface buoys and the underwater sensors, to conserve energy, and allow all data to be transmitted when close by. When an UAV is close to a sensor, it would use the JANUS protocol for communication, which has a method for "waking up" sleeping sensors and announcing it is close. The JANUS protocol uses acoustic communication, which is good underwater since it has a long range, however it has very low bit rates. This is okay for a single sensor sending data to a UAV, and since the UAV moves slowly, the underwater sensor can transmit data over a long period. UAVs periodically also returns closer to the surface buoys, and can then transfer the stored data from the sensors to the surface buoy which can send the data to the back-end through Argos. The communication between the UAV and the surface buoy will need higher data rates than JANUS allows, so the data can be sent using optical communication, which allows for higher data rates, but the UAV needs to be close. At the same time, the UAV can dock at the buoy to recharge.

The picture below is a rough illustration of how the system architecture would look.



The sensors will need to be battery powered, since they won't use any energy harvesting, but this should be manageable as our network architecture is energy efficient. The UAVs will also be battery powered, but be rechargeable from charging ports on the surface buoys. They will move slowly to be as energy efficient as possible. The surface buoys will require a lot of power, as they are responsible for being LoRa access points, charging the UAVs and receiving their data, and transmitting it to Argos. Since there isn't an energy grid that the surface buoy can connect to in the sea between Svalbard and the North Pole, it will need to power itself through some other way. The most reliable way this far north would be to have a fuel tank attached and a motor that can power the equipment.

Solution example 3:

This feels like a non-critical system where we want to ensure that our sensors last for a long time without maintenance. It is clear that we need some kind of sensors for the temperature and salinity, however it is a bit more unclear what "collect data of the ice cap" means, but assume that this data is images.

As for the sensors, I would recommend a system where I take inspiration from LoRa except modified for aquatic use, by using acoustics to send data from the sensors to a buoy. The buoy will store data until an aircraft passes (in a specified time slot). The aircraft picks up the data and takes pictures of the ice cap. This solution may be a bit of an overkill, for example when one buoy is deployed at Svalbard, an aircraft would be expensive. However, if the solution is deployed around all of the Arctic, this would be more cost effective. Also, it will also be the solution for the collection of "data of the ice cap" as specified.

As for more details, the senors will be low powered units, attached to the ocean floor, which harvest some energy from tidal forces and waves. The sensors will transmit the data acoustically in a given time slot to the buoy, perhaps using JANUS. The buoy will respond to sync the sensors' clocks, keeping them inline. The buoys can be solar powered for the summer months, and operate on battery during the winter. If not placed in a location that freezes, the buoy can harvest energy from waves. Also, the buoy will store the data and transmit it later to the airplane, making this a delayed forwarding network. If this delay would not suffice, the buoys can transmit via radio (a cheap alternative) to passing ships or use Starlink/Argos/SCaN depending on the amount of data collected (a for sure more expensive alternative)

Solution example 4: (just lacking a discussion about the very limited data rates for Argos)

This monitoring data will be in the shape of simple sensor data and is non-critical. Since the solution would most likely have to be either powered by batteries or some form of energy harvesting solution, it is resource-constrained.

There will be large beacons with built in wave energy harvesting solutions and large batteries to ensure they have constant power. These beacons will also have sonar on

them that can track the contours of "nearby" (still pretty far away) ice and rods extending deep into the ocean that can track both temperature and salinity at different depths. There will also be GPS modules on these beacons to ensure we can keep track of them, if they for some reason move.

There will also be one AUV with its own temperature, salinity and depth sensors for every beacon. These AUVs will go as far away from their beacon as their battery allows (including the route back plus margins, of course) and track salinity and temperature at different depths. Since GPS does not perform well underwater, these AUVs will also have to have some sort of motion sensors that can estimate where they've been based on the motions originating from the beacon (start position). When an AUV has finished its trip and gone back to the beacon, it will stay there for however long is necessary to let the AUV fully recharge again and upload its sensor data to the beacon. If the battery in the beacon the AUV is in on a low energy level, the AUV will simply wait for longer so the beacon can get energy back via its wave energy harvesting solution. There will be some sort of physical coupling between the AUV and beacon for it to charge efficiently. Since the AUV will be in a fixed and predictable position, we can use lasers to send all the data from the AUV to the beacon. This laser link will be half-duplex, so it can also be used for the beacon to send ACKs back to the AUV.

All of the beacon and AUV sensor data will be sent via the beacon to the NIPR cloud servers via Argos, since it is unlikely any other technology would give us consistent coverage around the entire North Pole (since Svalbard is merely a test site, we need to ensure it scales beyond this area).